Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

There are 4 bags with 6 apples each. How many apples? Student's answer: 10

Find the total.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 4 and 6. The bags are equal groups, so multiply 4 × 6 = 24.	**24 apples**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

There are 3 rows with 8 chairs each. How many chairs? Student's answer: 11

Find the total.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added instead of multiplying. 3 rows of 8 is 3 × 8 = 24.	**24 chairs**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 5 groups of 7 as a multiplication equation. Student writes: 5 + 7

Write the equation.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student wrote an addition. Five groups of seven is 5 × 7.	**5 × 7 = 35**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

There are 6 groups of 4 counters. Student's answer: 6 × 6 = 36

Write and solve the product.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 6 for the number in each group. Each group has 4 counters, so 6 × 4 = 24.	**6 × 4 = 24**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says $9 \times 0 = 9$.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Any number times 0 is 0, because there are no items in the groups.	**No. $9 \times 0 = 0$.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 2 groups of 5 is 2 × 5 = 7.

Is this right?

What did the student do wrong? Why do you think the student made this error?

The student added instead of multiplying. 2 × 5 = 10.

Solve the problem correctly. Show all of your work.

No. 2 × 5 = 10.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

24 ÷ 6 = ? Student's answer: 6

Divide.

What did the student do wrong? Why do you think the student made this error?

The student gave the divisor, not the quotient. 6 × 4 = 24, so 24 ÷ 6 = 4.

Solve the problem correctly. Show all of your work.

4

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

18 stickers are put into groups of 3. How many groups? Student's answer: 15

Find the number of groups.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 3 from 18. Divide 18 by 3 to find the groups: 6.	**6 groups**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Share 20 marbles equally among 4 kids. How many each? Student's answer: 16

Share equally.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 4 from 20. Divide 20 by 4 to share equally: 5 each.	**5 marbles each**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

35 ÷ 5 = ? Student's answer: 40

Divide.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied instead of dividing. 5 × 7 = 35, so 35 ÷ 5 = 7.	**7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many groups of 4 are in 28? Student's answer: 24

Find the number of groups.

What did the student do wrong? Why do you think the student made this error?

The student gave 4 × 6, not the number of groups. 28 ÷ 4 = 7.

Solve the problem correctly. Show all of your work.

7 groups

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Share 12 pencils into 3 equal groups. Student's answer: 3 each

Share equally.

What did the student do wrong? Why do you think the student made this error?

The student gave the number of groups, not the size of each group. 12 ÷ 3 = 4 each.

Solve the problem correctly. Show all of your work.

4 pencils each

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box has 8 crayons. There are 6 boxes. How many crayons? Student's answer: 14

Solve the word problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 8 and 6. There are 6 equal boxes, so multiply 6 × 8 = 48.	**48 crayons**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A bus carries 36 students shared equally among 4 vans. How many per van? Student's answer: 40

Solve the word problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 36 and 4. The students are shared equally, so divide 36 ÷ 4 = 9.	**9 students per van**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A pack has 9 cards. Lisa has 5 packs. How many cards? Student's answer: 14

Solve the word problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added. Five equal packs means multiplying 9 × 5 = 45.	**45 cards**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A class has 24 students in 6 equal rows. How many per row? Student's answer: 30

Solve the word problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 24 + 6. The students are shared into equal rows, so divide 24 ÷ 6 = 4.	**4 students per row**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

There are 7 bags with 5 apples each. How many apples? Student's answer: 12

Solve the word problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added. Seven equal bags means 7 × 5 = 35.	**35 apples**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

48 pencils are packed into boxes of 8. How many boxes? Student's answer: 56

Solve the word problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied, but the pencils are being separated into boxes. 48 ÷ 8 = 6.	**6 boxes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

? × 4 = 28. Student's answer: 24

Find the unknown number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student found 4 × 6 but not 28. The unknown factor is 28 ÷ 4 = 7.	**7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$6 \times ? = 42$. Student's answer: 36

Find the unknown number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 6×6 instead. The unknown factor is $42 \div 6 = 7$.	**7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

35 ÷ ? = 5. Student's answer: 30

Find the unknown number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted incorrectly. 5 × 7 = 35, so the unknown number is 7.	**7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

? ÷ 3 = 9. Student's answer: 3

Find the unknown number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided the wrong way. 9 × 3 = 27, so the unknown number is 27.	**27**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$9 \times ? = 63$. Student's answer: 54

Find the unknown number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 9×6. The unknown factor is $63 \div 9 = 7$.	**7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

? × 6 = 54. Student's answer: 48

Find the unknown number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 8 × 6. The unknown factor is 54 ÷ 6 = 9.	**9**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 3 × 4 is not the same as 4 × 3. Is this right?

Check the commutative property.

What did the student do wrong? Why do you think the student made this error?

The student misused the commutative property. Changing the order of factors does not change the product.

Solve the problem correctly. Show all of your work.

No. Both equal 12.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student uses the distributive property: $7 \times 6 = 7 \times 5 + 1 = 36$. Is this right?

Check the strategy.

What did the student do wrong? Why do you think the student made this error?

The student used 1 instead of 7 for the second part. 6 = 5 + 1, so 7 × 1 = 7 and 35 + 7 = 42.

Solve the problem correctly. Show all of your work.

No. 7 × 6 = 42.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student solves $(2 \times 3) \times 4$ as $2 \times (3 + 4) = 14$. Is this right?

Check the grouping.

What did the student do wrong? Why do you think the student made this error?

The student changed × to + in the grouping. (2 × 3) × 4 = 6 × 4 = 24.

Solve the problem correctly. Show all of your work.

No. (2 × 3) × 4 = 24.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 6 × 1 = 1 × 6 = 0. Is this right?

Check the product.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a multiplication error. Any number times 1 is that number, so 6 × 1 = 6.	**No. 6 × 1 = 6.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student solves 4×9 as $4 \times 10 + 4 = 44$. Is this right?

Check the strategy.

What did the student do wrong? Why do you think the student made this error?

The student added 4 instead of subtracting it. $9 = 10 - 1$, so $4 \times 10 - 4 = 36$.

Solve the problem correctly. Show all of your work.

No. $4 \times 9 = 36$.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student solves $3 \times (4 + 2)$ as $3 \times 4 + 2 = 14$. Is this right?

Check the distributive property.

What did the student do wrong? Why do you think the student made this error?

The student multiplied only the first part. Multiply 3 by each addend: $3 \times 4 + 3 \times 2 = 18$.

Solve the problem correctly. Show all of your work.

No. $3 \times (4 + 2) = 18$.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

24 ÷ 8 = ? Student says: 4, because 8 + 4 = 12.

Divide.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used addition. Division asks what times 8 is 24, and 8 × 3 = 24.	**3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

? ÷ 4 = 6. Student's answer: 10

Find the unknown dividend.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 4 and 6. The unknown number is 6 × 4 = 24.	**24**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

6 × ? = 30. Student's answer: 36

Find the unknown factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 6 × 6. The unknown factor is 30 ÷ 6 = 5.	**5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

18 ÷ 2 = ? Student's answer: 16

Divide.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 2. 2 × 9 = 18, so 18 ÷ 2 = 9.	**9**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

21 ÷ 3 = ? Student's answer: 6

Divide.

What did the student do wrong? Why do you think the student made this error?

The student gave a number one less than the answer. 3 × 7 = 21, so 21 ÷ 3 = 7.

Solve the problem correctly. Show all of your work.

7

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says $0 \div 5 = 5$.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Zero divided by any nonzero number is zero, because there is nothing to share.	**No. $0 \div 5 = 0$.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

7 × 8 = ? Student's answer: 54

Multiply within 100.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a multiplication error. 7 × 8 = 56.	**56**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

6 × 9 = ? Student's answer: 45

Multiply within 100.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 5 × 9. 6 × 9 = 54.	**54**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

56 ÷ 7 = ? Student's answer: 6

Divide within 100.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a division error. 7 × 8 = 56, so 56 ÷ 7 = 8.	**8**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$4 \times 7 = ?$ Student's answer: 21

Multiply within 100.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 3 × 7. 4 × 7 = 28.	**28**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

81 ÷ 9 = ? Student's answer: 8

Divide within 100.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave one less than the answer. 9 × 9 = 81, so 81 ÷ 9 = 9.	**9**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

5 × 6 = ? Student's answer: 35

Multiply within 100.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 5 × 7. 5 × 6 = 30.	**30**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A store has 5 boxes with 8 pencils each, then sells 6 pencils. How many are left? Student's answer: 34

Solve the two-step problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted wrong. 5 × 8 = 40 and 40 + 0 is not what happens here. Add all the pencils first, then take away the sold ones, 40 - 6 = 34.	**34 pencils (5 × 8 = 40, then 40 - 6)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Jake buys 4 packs of 6 gum and 3 extra pieces. How many pieces in all? Student's answer: 13

Solve the two-step problem.

What did the student do wrong? Why do you think the student made this error?

The student added the pack number instead of multiplying. 4 × 6 = 24, and 24 + 3 = 27.

Solve the problem correctly. Show all of your work.

27 pieces

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

es are split evenly into 3 bags. Then 2 more apples go into each bag. How many in each bag? Student's ar

Solve the two-step problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student left out the 2 extra apples. 24 ÷ 3 = 8, then 8 + 2 = 10.	**10 apples**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

8 students are in 4 equal groups. Then 3 students join group 1. How many are in group 1? Student's answer:

Solve the two-step problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student left out the 3 students who joined. 28 ÷ 4 = 7, then 7 + 3 = 10.	**10 students**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Estimate 49 × 6. Student's answer: about 250

Estimate the product.

What did the student do wrong? Why do you think the student made this error?

The student rounded 49 down too far. Round 49 to 50, then 50 × 6 = 300.

Solve the problem correctly. Show all of your work.

About 300

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Ana has $20 and buys 3 books for $5 each. How much money is left? Student's answer: $35

Solve the two-step problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the cost to the money. **3 × $5 = $15, and $20 - $15 = $5.**	**$5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Pattern: 4, 8, 12, 16. Student's next number: 18

Find the next number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 2 instead of 4. The pattern adds 4 each time, so the next number is 20.	**20**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Pattern: 3, 6, 9, ... Student's next number: 13

Find the next number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 4 instead of 3. The pattern adds 3, so the next number is 12.	**12**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$2 \times 5 = 10$, $4 \times 5 = 20$, $6 \times 5 = 30$. Student says the next product is $8 \times 5 = 45$.

Find the next product.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student solved 8×5 wrong. The pattern adds 10 each time, and $8 \times 5 = 40$.	**40**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Pattern: 5, 10, 15, 25. Student says the pattern is correct. Is it?

Check the pattern.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The pattern adds 5 each time, so 20 is missing between 15 and 25.	**No. The next number after 15 is 20.**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Pattern: 1, 4, 7, 10, 13. Student's next number: 15

Find the next number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 2 instead of 3. The pattern adds 3, so the next number is 16.	**16**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student lists 9 × 3 = 27, 9 × 4 = 36, and says 9 × 5 = 40. Is this right?

Check the product.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 4 instead of 9. 9 × 5 = 45.	**No. 9 × 5 = 45.**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 46 to the nearest ten. Student's answer: 40

Round to the nearest ten.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student rounded down. 46 is closer to 50 than to 40.	**50**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 268 to the nearest hundred. Student's answer: 200

Round to the nearest hundred.

What did the student do wrong? Why do you think the student made this error?

The student rounded down. 268 is closer to 300 than to 200.

Solve the problem correctly. Show all of your work.

300

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 85 to the nearest ten. Student's answer: 80

Round to the nearest ten.

What did the student do wrong? Why do you think the student made this error?

The student rounded down. 85 rounds up to 90.

Solve the problem correctly. Show all of your work.

90

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 150 to the nearest hundred. Student's answer: 100

Round to the nearest hundred.

What did the student do wrong? Why do you think the student made this error?

The student rounded down. 150 is halfway, and the rule is to round up to 200.

Solve the problem correctly. Show all of your work.

200

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 77 to the nearest ten. Student's answer: 70

Round to the nearest ten.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student rounded down. 77 is closer to 80.	**80**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 612 to the nearest ten. Student's answer: 600

Round to the nearest ten.

What did the student do wrong? Why do you think the student made this error?

The student rounded to the nearest hundred. 612 rounds to 610 at the tens place.

Solve the problem correctly. Show all of your work.

610

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

245 + 128 = ? Student's answer: 363

Add within 1,000.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made an addition error. 245 + 128 = 373.	**373**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

500 - 275 = ? Student's answer: 325

Subtract within 1,000.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a subtraction error. 500 - 275 = 225.	**225**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

330 + 470 = ? Student's answer: 700

Add within 1,000.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made an addition error. 330 + 470 = 800.	**800**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

612 - 348 = ? Student's answer: 364

Subtract within 1,000.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a subtraction error. 612 - 348 = 264.	**264**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

199 + 201 = ? Student's answer: 300

Add within 1,000.

What did the student do wrong? Why do you think the student made this error?

The student made an addition error. 199 + 201 = 400.

Solve the problem correctly. Show all of your work.

400

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

846 - 90 = ? Student's answer: 856

Subtract within 1,000.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 90 instead of subtracting. 846 - 90 = 756.	**756**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

3 × 70 = ? Student's answer: 2,100

Multiply.

What did the student do wrong? Why do you think the student made this error?

The student put the zeros in the wrong place. 3 × 7 = 21, so 3 × 70 = 210.

Solve the problem correctly. Show all of your work.

210

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

5 × 30 = ? Student's answer: 35

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot the zero. 5 × 3 = 15, so 5 × 30 = 150.	**150**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

8 × 40 = ? Student's answer: 3,200

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added an extra zero. 8 × 4 = 32, so 8 × 40 = 320.	**320**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

6 × 90 = ? Student's answer: 504

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied incorrectly. 6 × 9 = 54, so 6 × 90 = 540.	**540**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

2 × 200 = ? Student's answer: 40

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student dropped the zeros. 2 × 2 = 4, so 2 × 200 = 400.	**400**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

7 × 50 = ? Student's answer: 320

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied incorrectly. 7 × 5 = 35, so 7 × 50 = 350.	**350**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 1/4 means 4 parts out of 1. Is this right?

Explain the fraction.

What did the student do wrong? Why do you think the student made this error?

The student reversed the meaning. The bottom number is the number of equal parts in the whole, and 1/4 means 1 of 4 equal parts.

Solve the problem correctly. Show all of your work.

No. 1/4 is 1 of 4 equal parts.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 1/3 of a shape means 3 parts are shaded. Is this right?

Explain the fraction.

What did the student do wrong? Why do you think the student made this error?

The student mixed up the numerator. 1/3 means 1 of the 3 equal parts is shaded.

Solve the problem correctly. Show all of your work.

No. 1/3 means 1 part out of 3 equal parts.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says the top number of a fraction is the number of equal parts. Is this right?

Check the parts of a fraction.

What did the student do wrong? Why do you think the student made this error?

The bottom number tells how many equal parts the whole has. The top number tells how many parts are counted.

Solve the problem correctly. Show all of your work.

No. The bottom number is the number of equal parts.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A shape is split into 8 equal parts with 1 shaded. Student writes 8/1.

Write the fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student reversed the numbers. 1 shaded part out of 8 equal parts is 1/8.	**1/8**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 1/4 is greater than 1/2 because 4 is bigger than 2. Is this right?

Compare the fractions.

What did the student do wrong? Why do you think the student made this error?

The student compared the denominators only. Halves are bigger pieces than fourths, so 1/2 is greater.

Solve the problem correctly. Show all of your work.

No. 1/2 is greater than 1/4.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A bar is split into 3 equal parts and 2 are shaded. Student writes 1/3.

Write the fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted only 1 part. 2 of 3 equal parts are shaded, so the fraction is 2/3.	**2/3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Place 1/2 on a number line from 0 to 1. Student places it at 1/4.

Place the fraction.

What did the student do wrong? Why do you think the student made this error?

The student placed the fraction at a quarter. 1/2 is halfway between 0 and 1.

Solve the problem correctly. Show all of your work.

Halfway between 0 and 1

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A number line from 0 to 1 has 4 equal parts. Student marks 3/4 at the 4th mark.

Place the fraction.

What did the student do wrong? Why do you think the student made this error?

The student marked the endpoint. 3/4 is the third mark from 0.

Solve the problem correctly. Show all of your work.

The 3rd mark from 0

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 1/4 and 2/4 are at the same point on a number line.

Is this right?

What did the student do wrong? Why do you think the student made this error?

The student thought the two fractions were the same point. 2/4 is twice as far from 0 as 1/4.

Solve the problem correctly. Show all of your work.

No. 1/4 and 2/4 are different points.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 1 and 2/2 are different points on a number line.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not see that 2/2 equals 1 whole. They are the same point.	**No. 2/2 equals 1, so they are the same point.**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 0 cannot be written as a fraction on a number line.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student missed that 0 can be written as 0/1 or 0/2, for example.	**No. 0 can be written as 0/1.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

On a number line from 0 to 2 with halves, student says 3/2 is at 2.

Place the fraction.

What did the student do wrong? Why do you think the student made this error?

The student placed 3/2 at the endpoint. 3/2 equals 1 1/2, which is halfway between 1 and 2.

Solve the problem correctly. Show all of your work.

3/2 is halfway between 1 and 2.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 1/2 equal to 2/4? Student's answer: no

Are the fractions equal?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not see that the fractions are equivalent. 2/4 is the same amount as 1/2.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write a fraction equal to 2/3 with a denominator of 6. Student's answer: 4/3

Write an equivalent fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student changed only the numerator. Multiply both the top and bottom by 2 to get 4/6.	**4/6**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 3/4 and 3/8. Student writes: 3/4 < 3/8

Compare the fractions.

What did the student do wrong? Why do you think the student made this error?

The student reversed the comparison. 3/4 uses bigger pieces, so 3/4 is greater.

Solve the problem correctly. Show all of your work.

3/4 > 3/8

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 4/4 = 2.

Is this right?

What did the student do wrong? Why do you think the student made this error?

The student made an error with the whole. 4/4 is 4 parts out of 4, which equals 1 whole.

Solve the problem correctly. Show all of your work.

No. 4/4 = 1.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 2/5 equal to 4/10? Student's answer: no

Are the fractions equal?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not check that both are the same amount. 2/5 = 4/10.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 3/6 = 1/3. Is this right?

Check the fractions.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student is right, but the reason needs checking. 3/6 simplifies to 1/2, not 1/3.	**No. 3/6 = 1/2.**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A task starts at 1:20 and ends at 2:05. How many minutes? Student's answer: 85

Find the elapsed time.

What did the student do wrong? Why do you think the student made this error?

The student added the minutes. From 1:20 to 2:00 is 40 minutes, then 5 more is 45.

Solve the problem correctly. Show all of your work.

45 minutes

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A class starts at 9:30 and ends at 10:00. How many minutes? Student's answer: 70

Find the elapsed time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added wrong. From 9:30 to 10:00 is 30 minutes.	**30 minutes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Start 4:10. End 4:45. How many minutes? Student's answer: 45

Find the elapsed time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted the end minutes only. From 4:10 to 4:45 is 35 minutes.	**35 minutes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Start 7:50. End 8:20. How many minutes? Student's answer: 20

Find the elapsed time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not count through the hour. From 7:50 to 8:00 is 10 minutes, then 20 more, for 30 minutes.	**30 minutes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Start 5:15. End 6:00. How many minutes? Student's answer: 75

Find the elapsed time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the wrong count. From 5:15 to 6:00 is 45 minutes.	**45 minutes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many minutes are in 2 hours? Student's answer: 100

Convert hours to minutes.

What did the student do wrong? Why do you think the student made this error?

The student counted hours as tens. Each hour has 60 minutes, so 2 hours is 120 minutes.

Solve the problem correctly. Show all of your work.

120 minutes

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A bottle holds 2 liters. Student's answer: 2 milliliters

Choose the unit.

What did the student do wrong? Why do you think the student made this error?

The student used the smallest unit. 2 liters is 2,000 milliliters.

Solve the problem correctly. Show all of your work.

2,000 milliliters

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A bag of rice weighs 1 kilogram. Student's answer: 1 gram

Choose the unit.

What did the student do wrong? Why do you think the student made this error?

The student used the smallest mass unit. 1 kilogram is 1,000 grams.

Solve the problem correctly. Show all of your work.

1,000 grams

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A paperclip weighs about 1 gram. Student's answer: 1 kilogram

Choose the unit.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used a much bigger unit. A paperclip weighs about 1 gram, which is far less than a kilogram.	**1 gram**

100

Find the Error

3.MD.2

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A pitcher holds 4 liters. Student's answer: 40 milliliters

Convert the unit.

What did the student do wrong? Why do you think the student made this error?

The student made a conversion error. 4 liters is 4,000 milliliters.

Solve the problem correctly. Show all of your work.

4,000 milliliters

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box of cereal weighs 500 grams. Student's answer: 5 kilograms

Convert the unit.

What did the student do wrong? Why do you think the student made this error?

The student made a conversion error. 500 grams is 0.5 kilograms, which is less than 1 kilogram.

Solve the problem correctly. Show all of your work.

0.5 kilograms

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A glass holds 250 milliliters. Student's answer: 250 liters

Choose the unit.

What did the student do wrong? Why do you think the student made this error?

The student used a much bigger unit. A glass holds about 250 milliliters, which is a quarter of a liter.

Solve the problem correctly. Show all of your work.

250 milliliters

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A picture graph: each picture stands for 5 kids. There are 6 pictures. Student's answer: 6 kids

Read the picture graph.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted the pictures, not the kids. 6 pictures times 5 is 30.	**30 kids**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A bar graph has a scale of 2. The bar reaches 4 marks. Student's answer: 4

Read the bar graph.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student read the marks, not the value. Each mark is worth 2, so 4 marks is 8.	**8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

dogs have 7 pictures and cats have 4 pictures. Each picture is 2 pets. How many more dogs than cats? Stu

Find the difference.

What did the student do wrong? Why do you think the student made this error?

The student subtracted the pictures and not the pets. Dogs have 14 and cats have 8, so the difference is 6.

Solve the problem correctly. Show all of your work.

6 more

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a bar graph's scale should always be 1.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The scale should match the data. A scale of 1 may make the graph too large to read.	**No. Choose a scale that fits the data.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ture graph shows 3 pictures and 4 pictures. Each picture is 10 items. How many items in all? Student's answ

Find the total.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted one picture only. 3 + 4 = 7 pictures, and 7 times 10 is 70.	**70 items**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a bar graph does not need a title.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A title tells the reader what the graph shows, so it is needed.	**No. A graph needs a title.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line plot shows 2 X's at 1/4 inch and 3 X's at 1/2 inch. How many X's in all? Student's answer: 2

Count the X's.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted only the X's at 1/4. There are 2 + 3 = 5 X's.	**5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line plot has 4 X's at 1/2 inch. Student says the total length is 4 inches.

Find the total length.

What did the student do wrong? Why do you think the student made this error?

The student multiplied wrong. 4 × 1/2 = 2 inches.

Solve the problem correctly. Show all of your work.

2 inches

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 1/2 inch and 1/4 inch are the same length.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student thought they were the same. 1/2 inch is twice as long as 1/4 inch.	**No. 1/2 is longer than 1/4.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a line plot must use only whole numbers.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Line plots can use fractions, such as 1/4 and 1/2.	**No. Line plots can use fractions.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line plot shows 5 X's at 1/2 and 3 X's at 1. How many measurements in all? Student's answer: 5

Count the measurements.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted only the X's at 1/2. 5 + 3 = 8 measurements.	**8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 6 X's at 1/2 inch equal 6 inches in total.

Find the total length.

What did the student do wrong? Why do you think the student made this error?

The student did not multiply by the length. 6 × 1/2 = 3 inches.

Solve the problem correctly. Show all of your work.

3 inches

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle is covered by 12 unit squares with no gaps. Student's answer: 14 square units

Find the area.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted wrong. The rectangle is covered by 12 unit squares.	**12 square units**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says the area of a 3 by 4 rectangle is 7 because 3 + 4 = 7.

Find the area.

What did the student do wrong? Why do you think the student made this error?

The student added the sides. Area is found by multiplying: 3 × 4 = 12 square units.

Solve the problem correctly. Show all of your work.

12 square units

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says area and perimeter are the same measurement.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Area measures the space inside a shape. Perimeter measures the distance around it.	**No. Area and perimeter are different.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 5 by 2 rectangle. Student says the area is 14 square units.

Find the area.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the sides to get perimeter. Area is 5 × 2 = 10 square units.	**10 square units**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A square with side length 6 units. Student's answer: 24 square units

Find the area.

What did the student do wrong? Why do you think the student made this error?

The student added the sides. Area is 6 × 6 = 36 square units.

Solve the problem correctly. Show all of your work.

36 square units

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 4 by 3 rectangle. Student's answer: 7 square units

Find the area.

What did the student do wrong? Why do you think the student made this error?

The student added the sides instead of multiplying. 4 × 3 = 12 square units.

Solve the problem correctly. Show all of your work.

12 square units

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 3 by 3 region is covered by 9 square tiles. Student says the area is 6 square units.

Count the square units.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student missed some squares. The region has 9 square tiles.	**9 square units**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle covers 20 squares that are 1 cm each. Student says 2 square cm.

Find the area.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted the rows. The rectangle covers 20 square centimeters.	**20 square cm**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student measures a rectangle's area with a ruler: 5 cm, and says 5 square cm.

Find the area.

What did the student do wrong? Why do you think the student made this error?

A ruler measures length, not area. Area must be counted with square units.

Solve the problem correctly. Show all of your work.

Count the squares that cover the shape.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A shape covers 15 tiles. Student says the area is 15 units long.

What is the area?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the length unit. Area is measured in square units: 15 square units.	**15 square units**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 2 by 5 rectangle is covered by squares. Student says 7 square units.

Find the area.

What did the student do wrong? Why do you think the student made this error?

The student added the sides. The area is 2 × 5 = 10 square units.

Solve the problem correctly. Show all of your work.

10 square units

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle covers 6 squares that are each 2 cm on a side. Student says 6 cm.

What is the area?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used length instead of area. The area is 6 square units, and each square is 2 cm by 2 cm.	**6 square units (each 2 cm by 2 cm)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a rectangle with sides 6 and 4. Student's answer: 10

Find the area.

What did the student do wrong? Why do you think the student made this error?

The student added the sides. Area is 6 × 4 = 24.

Solve the problem correctly. Show all of your work.

24 square units

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a rectangle 7 by 3. Student says: 10 square units because 7 + 3 = 10.

Find the area.

What did the student do wrong? Why do you think the student made this error?

The student added the sides. Area is 7 × 3 = 21.

Solve the problem correctly. Show all of your work.

21 square units

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a rectangle 9 by 2. Student's answer: 18 meters

Find the area.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the wrong unit. Area is measured in square meters.	**18 square meters**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a 5 by 5 square. Student's answer: 20

Find the area.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the sides. Area is 5 × 5 = 25.	**25 square units**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a rectangle 8 by 6. Student's answer: 14

Find the area.

What did the student do wrong? Why do you think the student made this error?

The student added the sides. Area is 8 × 6 = 48.

Solve the problem correctly. Show all of your work.

48 square units

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle has area 24 and one side is 6. Student says the other side is 18.

Find the missing side.

What did the student do wrong? Why do you think the student made this error?

The student subtracted instead of dividing. Area = side × side, so 24 ÷ 6 = 4.

Solve the problem correctly. Show all of your work.

4

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the perimeter of a rectangle 5 by 3. Student's answer: 15

Find the perimeter.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added only 2 sides. The perimeter adds all 4 sides: 5 + 3 + 5 + 3 = 16.	**16**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the perimeter of a square with side 4. Student's answer: 8

Find the perimeter.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added only 2 sides. The perimeter is 4 + 4 + 4 + 4 = 16.	**16**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle has sides 6, 2, 6, and 2. Student's answer: 8

Find the perimeter.

What did the student do wrong? Why do you think the student made this error?

The student added only 2 sides. The perimeter is 6 + 2 + 6 + 2 = 16.

Solve the problem correctly. Show all of your work.

16

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has sides 3, 4, and 5. Student's answer: 7

Find the perimeter.

What did the student do wrong? Why do you think the student made this error?

The student left out one side. The perimeter is 3 + 4 + 5 = 12.

Solve the problem correctly. Show all of your work.

12

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle is 10 by 4. Student's answer: 40

Find the perimeter.

What did the student do wrong? Why do you think the student made this error?

The student multiplied the sides instead of adding them. The perimeter is 10 + 4 + 10 + 4 = 28.

Solve the problem correctly. Show all of your work.

28

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle has perimeter 30 and one side is 8. Student says the other side is 11.

Find the missing side.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not account for two sides. The two 8s use 16, so 30 - 16 = 14 for the other two sides, and each is 7.	**7**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a square is not a rectangle. Is this right?

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A square has 4 right angles and two pairs of parallel sides, so it is a rectangle.	**No. A square is a rectangle.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a rhombus is always a square. Is this right?

Is this right?

What did the student do wrong? Why do you think the student made this error?

A rhombus has 4 equal sides, but it needs right angles to be a square.

Solve the problem correctly. Show all of your work.

No. A rhombus is not always a square.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a trapezoid is a parallelogram. Is this right?

Is this right?

What did the student do wrong? Why do you think the student made this error?

A trapezoid has only one pair of parallel sides. A parallelogram has two pairs.

Solve the problem correctly. Show all of your work.

No. A trapezoid is not a parallelogram.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Name a shape with 4 right angles. Student's answer: rhombus

Name the shape.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A rhombus does not have right angles. A rectangle has 4 right angles.	**Rectangle (or square)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a parallelogram has 4 right angles.

Is this right?

What did the student do wrong? Why do you think the student made this error?

A parallelogram has opposite sides parallel, but its angles do not have to be right angles.

Solve the problem correctly. Show all of your work.

No. A parallelogram has opposite sides that are parallel.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a quadrilateral must have 4 equal sides.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A quadrilateral has 4 sides, but they do not have to be equal.	**No. A quadrilateral has 4 sides, not necessarily equal.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle is split into 4 parts that are not equal in area. Student says each part is 1/4.

Are the parts equal?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The parts are not equal in area, so each part is not 1/4 of the whole.	**No. The parts must have equal area to be fourths.**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A shape is split into 4 equal parts. Student says each part is 1/2.

Name the fraction.

What did the student do wrong? Why do you think the student made this error?

The student used the wrong denominator. 4 equal parts make fourths, so each part is 1/4.

Solve the problem correctly. Show all of your work.

1/4

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle is split into 3 equal parts. Student says each part is 1/2.

Name the fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 2 instead of 3. 3 equal parts make thirds, so each part is 1/3.	**1/3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says equal parts must look exactly the same.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Equal parts need the same area, not the same shape.	**No. Equal parts need the same area.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A shape is split into 6 equal parts. Student says each part is 1/3.

Name the fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 3 instead of 6. 6 equal parts make sixths, so each part is 1/6.	**1/6**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 2/4 is the same as 1/4.

Is this right?

What did the student do wrong? Why do you think the student made this error?

The student did not compare the amounts. 2/4 is twice 1/4, and 2/4 equals 1/2.

Solve the problem correctly. Show all of your work.

No. 2/4 is not the same as 1/4.

Original task. Format inspired by error analysis routines.